Mandatory Assignment 1

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1 Exercise

1.1 Market excess returns

We download monthly data for the S&P 500 index (`^GSPC`) and the 13-week Treasury Bill rate (`^IRX`) from Yahoo! Finance for the period January 1, 2000 to February 28, 2026. The `^IRX` series is an annualized yield in percent, so it is converted into a monthly risk-free rate as

$$r_{f,t} = (1 + IRX_t/100)^{1/12} - 1.$$

Monthly excess market returns are then computed as

$$\tilde{r}_{m,t} = r_{m,t} - r_{f,t},$$

where $r_{m,t}$ denotes the monthly return on the S&P 500 index. We use the estimated monthly mean excess return μ_m and monthly variance σ_m^2 to calibrate the synthetic asset universe used in the remainder of the assignment.

Table 1: Estimated monthly and annualized market moments

	metric	value
0	mu_m	0.0045
1	sigma2_m	0.0019
2	mu_m_annualized	0.0545
3	sigma2_m_annualized	0.0228

Table Table 1 reports the estimated monthly mean excess market return and variance, together with their annualized counterparts. The estimated monthly mean excess return is approximately 0.45%, while the annualized mean excess return is approximately 5.45%. The corresponding annualized variance is approximately 0.0228.

1.2 Synthetic asset universe

Next, we construct a universe of $N = 50$ synthetic assets following the single-factor model

$$\tilde{r}_{i,t} = \beta_i \tilde{r}_{m,t} + \varepsilon_{i,t}.$$

The factor loadings β_i and idiosyncratic volatilities $\sigma_{\varepsilon,i}$ are drawn from the distributions specified in the assignment. Using the estimated market moments from Exercise 1, we construct the true mean vector μ and covariance matrix Σ, which are treated as the true parameters throughout the remainder of the assignment.

1.3 Functions for the efficient frontier and tangency portfolio

We next implement functions to compute the tangency portfolio and the mean-variance efficient frontier for a given mean vector μ and covariance matrix Σ. The tangency portfolio maximizes the Sharpe ratio, while the efficient frontier contains the portfolios with minimum variance for a given target return.

1.4 Efficient frontier based on the true parameters

Using the true parameter values μ and Σ, we compute the mean-variance efficient frontier and the corresponding tangency portfolio. The resulting frontier describes the best attainable risk-return combinations under the true data-generating process.

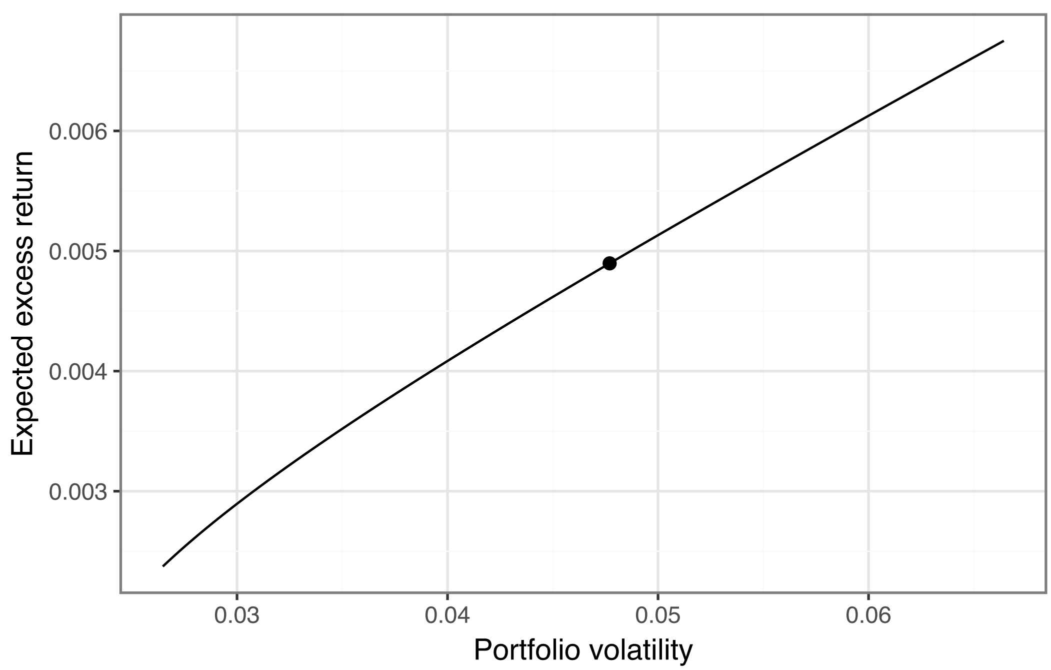

Figure 1: Mean-variance efficient frontier and tangency portfolio based on the true parameters.

Figure Figure 1 shows the efficient frontier implied by the true parameter values. The marked point denotes the tangency portfolio, which achieves the highest Sharpe ratio among all feasible portfolios.

1.5 Estimation error and the shape of the efficient frontier

To study estimation error, we simulate asset returns from the true data-generating process and estimate the mean vector $\hat{\mu}$ and covariance matrix $\hat{\Sigma}$ from the simulated sample. These sample estimates are then used to construct a plug-in efficient frontier.

We first compare the true efficient frontier with the frontier estimated from a single simulated sample with $T = 120$ monthly observations.

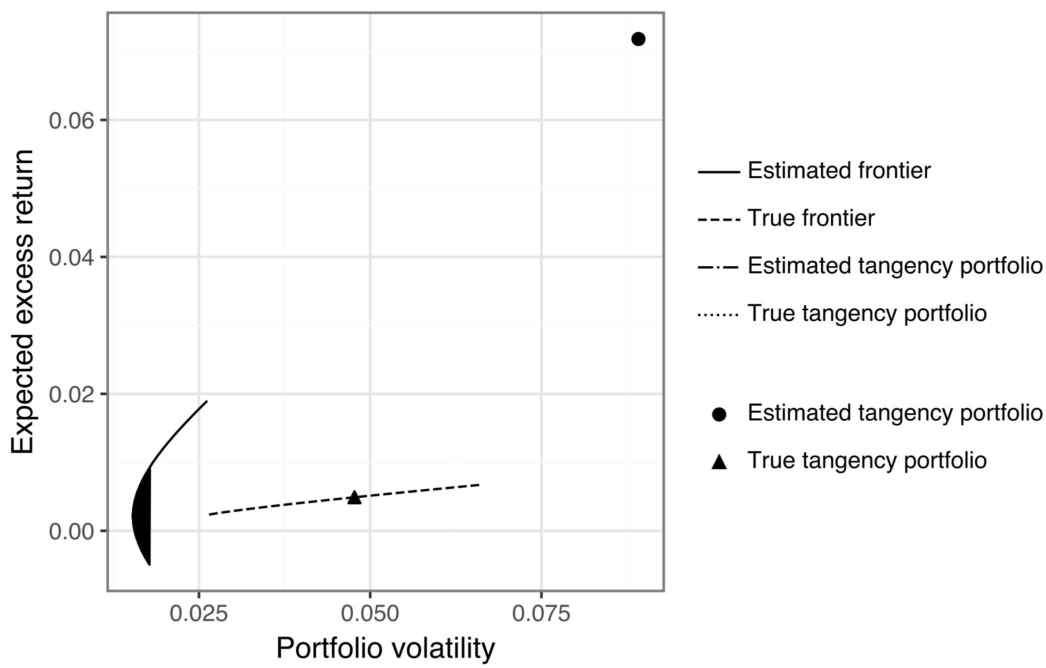

Figure 2: True and estimated efficient frontiers based on one simulated sample.

To study how sample size affects estimation error, we repeat the simulation for different values of T.

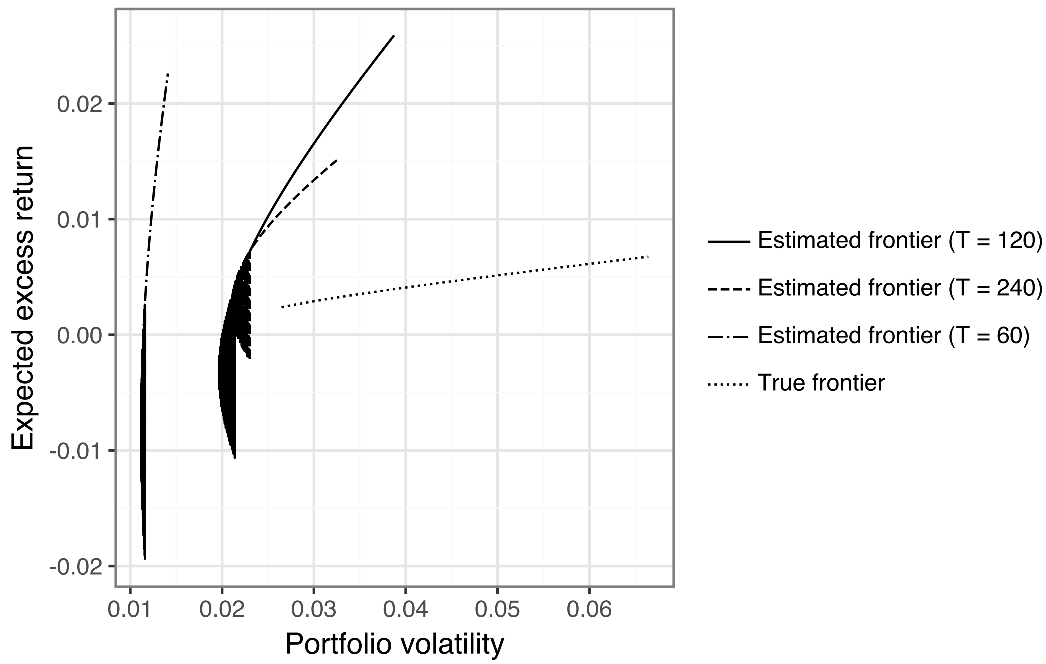

Figure 3: Estimated efficient frontiers for different sample sizes together with the true frontier.

Figure 2 and Figure 3 illustrate how estimation error affects the shape of the efficient frontier. When the sample size is small relative to the number of assets, the estimated frontier can deviate substantially from the true frontier. As T increases, the estimated frontier becomes more stable and approaches the true frontier.

1.6 Out-of-sample performance of the plug-in tangency portfolio

To evaluate the impact of estimation error, we compare the true tangency portfolio with the plug-in tangency portfolio. The true tangency portfolio is constructed using the true parameters μ and Σ, while the plug-in tangency portfolio is based on the estimated sample moments $\hat{\mu}$ and $\hat{\Sigma}$.

The tangency portfolio weights are given by

$$w_{tan} = \frac{\Sigma^{-1}\mu}{\iota^\top \Sigma^{-1}\mu},$$

where ι denotes a vector of ones.

The Sharpe ratio of a portfolio with weights w is defined as

$$SR = \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}}.$$

To analyze the effect of estimation error, asset returns are simulated from the multivariate normal distribution

$$r_t \sim \mathcal{N}(\mu, \Sigma).$$

For each simulated sample with size T, the sample estimates $\hat{\mu}$ and $\hat{\Sigma}$ are computed and used to construct the plug-in tangency portfolio. The resulting portfolio is then evaluated using the true parameters μ and Σ.

Table 2: Average Sharpe ratios of the true and plug-in tangency portfolios across sample sizes.

	T	true_sharpe	plugin_sharpe
0	60	0.102665	0.006628
1	120	0.102665	0.008607
2	240	0.102665	0.014701
3	480	0.102665	0.022470

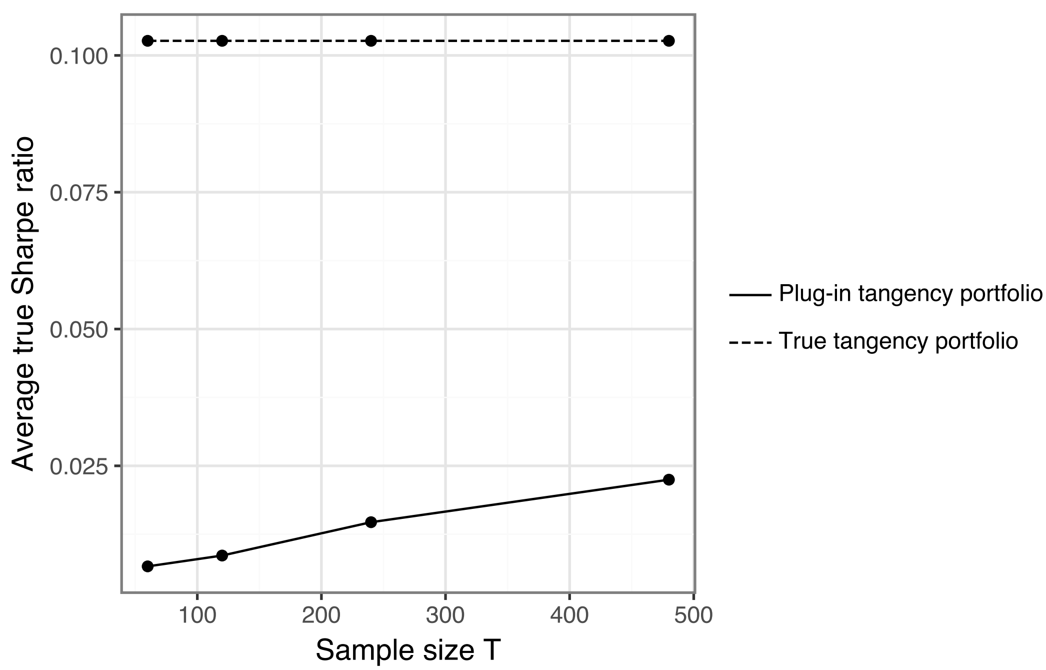

Figure 4: True and plug-in tangency portfolio performance evaluated under the true parameters.

Table 2 reports the average Sharpe ratios of the true tangency portfolio and the plug-in tangency portfolio for different sample sizes.

The results show that the plug-in tangency portfolio performs substantially worse than the true tangency portfolio when T is small. As T increases, the plug-in Sharpe ratio improves because the estimates $\hat{\mu}$ and $\hat{\Sigma}$ become more accurate.

Figure 4 illustrates the same pattern. The performance of the plug-in tangency portfolio increases with the sample size, but remains below the performance of the true tangency portfolio.

1.7 Alternative portfolio strategies

To study the robustness of different portfolio construction methods, we compare the plug-in tangency portfolio with several alternative strategies.

The first benchmark is the equally weighted portfolio, defined as

$$w_{EW} = \frac{1}{N}\iota.$$

This portfolio assigns equal weights to all assets and does not depend on estimated parameters.

The second strategy is the global minimum variance portfolio, which minimizes portfolio variance subject to the weights summing to one. The weights are given by

$$w_{GMV} = \frac{\Sigma^{-1}\iota}{\iota^\top \Sigma^{-1}\iota}.$$

This strategy depends only on the covariance matrix and does not require estimates of expected returns.

The third strategy is the inverse-volatility portfolio, where assets with higher volatility receive lower weights. The weights are defined as

$$w_i = \frac{1/\sigma_i}{\sum_{j=1}^{N} 1/\sigma_j}.$$

Table 3: Average true Sharpe ratios across portfolio strategies and sample sizes.

strategy T	Equally weighted	Global minimum variance	Inverse volatility	Plug-in tangency
60	0.1018	0.0219	0.102	0.0061
120	0.1018	0.0369	0.102	0.0099
240	0.1018	0.0433	0.102	0.0140
480	0.1018	0.0452	0.102	0.0235

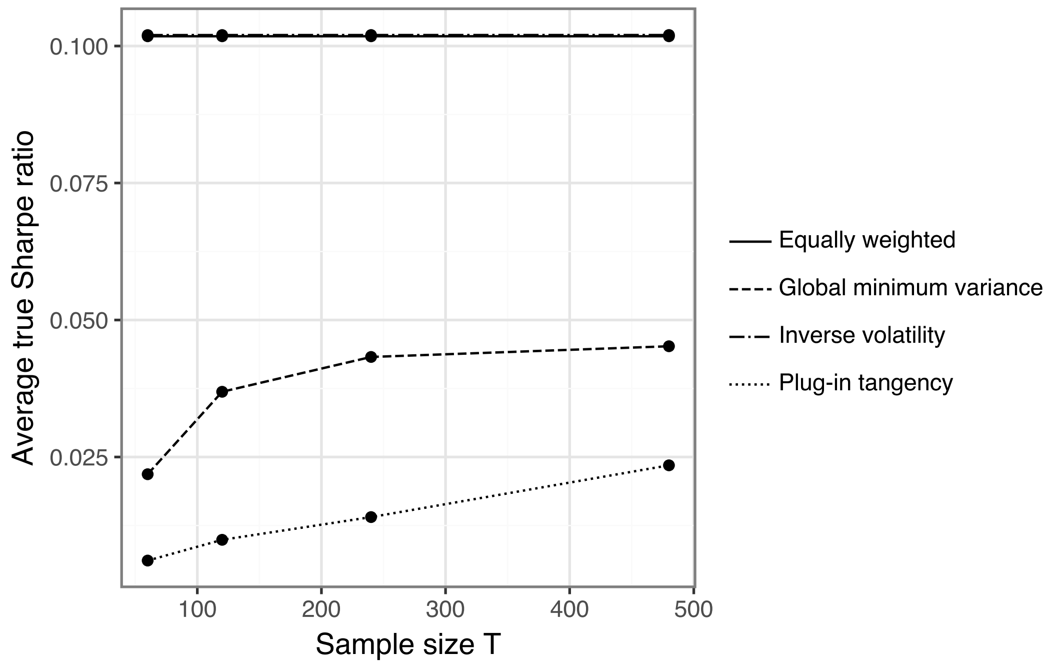

Figure 5: Average true Sharpe ratio across portfolio strategies.

Figure Figure 5 compares the average Sharpe ratios of the different strategies across sample sizes.

The results show that the plug-in tangency portfolio is the most sensitive to estimation error. In contrast, the equally weighted and inverse-volatility portfolios perform more consistently across different sample sizes because they rely less on estimated parameters. As the sample size increases, the performance of the plug-in tangency portfolio improves and gradually approaches the true optimum.